

FOL: A Functional Object Lisp

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Abstract

We present FOL (Functional Object Lisp), a Lisp dialect combining Clojure’s persistent data structures with CLOS-style object orientation and Dylan-inspired naming. We argue that persistent objects and CLOS are *synergistic*: their combination yields versioned objects via the MOP, declarative event sourcing via method combinations, observable slots via predicate dispatch over change events, and lazy schema evolution—patterns more natural than in either parent alone. Benchmarks show persistence adds only 1–3x overhead for sequential operations; the current 300–500x gap is attributable to interpretation. A preliminary transpiler to Common Lisp, with a single optimization pass (fixed-arity dispatch), reaches parity with hand-written code, confirming compilation as the primary optimization target. FOL is bootstrapped in Common Lisp atop FSet and Sycamore, providing Clojure-compatible sequence operations, transducers, and a meta-object protocol adapted for immutable storage.

CCS Concepts

• **Software and its engineering** → **Functional languages; Object oriented languages; Extensible languages; Language features.**

Keywords

Functional programming, object-oriented programming, Lisp, persistent data structures, CLOS, MOP, transducers

1 Introduction

Common Lisp [18] provides powerful object-oriented features through CLOS [2] and metaprogramming through the MOP [13]. Clojure [10] introduced persistent data structures emphasizing immutability. Dylan [16] explored an object-oriented Lisp with clear naming conventions. Each tradition offers distinct strengths: CLOS provides multiple dispatch, method combinations, and full metaobject protocol introspection; Clojure provides immutable data structures with $O(\log n)$ updates and structural sharing; Dylan provides clean naming and module semantics.

FOL (Functional Object Lisp) synthesizes these traditions. Our central claim is that persistent objects and CLOS are not merely compatible but *synergistic*—their combination yields patterns that are more natural and more concise than in either parent alone:

- **Versioned objects via persistence and the MOP:** Persistent snapshots with structural sharing make version chains cheap (Section 3.5), providing an implicit Memento pattern. Mutable CLOS requires explicit deep copies; Clojure has cheap snapshots but no metaclass protocol to encapsulate versioning as a reusable abstraction.

- **Event sourcing via method combinations:** `:around` methods declaratively intercept all mutations to build immutable event logs, while predicate dispatch routes commands by content (Section 5.2). FOL’s `:around` applies automatically to all methods; Clojure’s equivalent wrapper must be invoked explicitly.
- **Category-based predicate dispatch:** A specificity hierarchy (predicate $>$ type $>$ destructuring $>$ any) that automatically orders clauses across categories while preserving definition-order within (Section 4.3.2).
- **Lazy schema evolution:** Immutability ensures old instances are never corrupted by class redefinition, and functional updates transparently produce instances of the new schema (Section 5.4)—trading CLOS’s explicit migration hooks for corruption safety.

Our benchmarks (Section 6) show that persistence adds only 1–3x overhead for sequential operations; the current 300–500x overhead reflects interpretation cost. A preliminary transpiler with a single optimization—fixed-arity dispatch—reaches parity with hand-written Common Lisp (Section 6.6), confirming that compilation eliminates the dispatch overhead entirely.

2 Language Design

2.1 Core Philosophy

FOL’s design commitments:

- (1) **Immutability by default:** All objects and collections (except streams, atoms, etc.) are persistent—in the sense of Driscoll et al. [5], preserving previous versions after modification, *not* database persistence
- (2) **Structural sharing:** Updates create new versions sharing structure with old
- (3) **Object identity:** Distinguishes version identity (`eq`) from value equality (`=`)
- (4) **Generic functions:** Multiple dispatch over destructuring patterns
- (5) **Meta-object protocol:** Introspection and extension via adapted MOP protocols (Section 3.6)

On Identity and Equality: Following Hickey’s epochal time model [11], FOL simplifies Common Lisp’s four-level equality hierarchy (`eq`/`eq1`/`equal`/`equalp`) to two levels. `eq` tests *reference identity*—each functional update produces a new object, so (`eq` `alice` `older-alice`) returns false even when both represent “Alice.” FOL’s `=` tests *value equality* via generic dispatch: numbers compare with numeric coercion, persistent objects compare by structural comparison of storage maps (using `fset:equal?`), and strings compare by content—subsuming the roles of `eq1`, `equal`, and `equalp`. Two independently constructed objects with identical

slot values are = and may be used as map keys (hashing is based on storage structure, not reference identity).

2.2 Syntax and Readability

FOL adopts Clojure's reader syntax with Dylan's <type> naming:

```
[1 2 3]           ; Vectors
{:name "Alice" :age 30} ; Maps
#{1 2 3 4}        ; Sets

;; Destructuring with predicate tests and type constraints
(defn factorial
  ([[n (<= 1)]] 1)
  ([n] (* n (factorial (dec n)))))

(defn summarize [[:keys [(name <string>) (age <number>)]]]
  (str name " is " age " years old"))
```

2.3 Type System

FOL's type hierarchy (Figure 1) begins with <persistent-object>, from which most primitive, collection, and user types derive. Mutable types such as streams and atoms inherit from standard-object instead. Persistent primitive types wrap Common Lisp equivalents in a val slot, accessed through fol-val.

3 Architecture and Implementation

3.1 Bootstrap Implementation

The bootstrap totals approximately 18,000 lines of Common Lisp (excluding tests) with 6,333 passing tests. Table 1 shows the major components.

Table 1: Bootstrap Implementation Components

File	LOC	Purpose
eval	2,806	Evaluator and special forms
seqop	3,418	Sequence operations
standard-names	3,034	Standard environment
collection	1,404	Persistent collections
reader	1,012	FOL reader/parser
mop	1,854	Metaclass and MOP protocols
primitives	1,623	Primitive type wrappers
(others)	2,849	Strings, I/O, errors, etc.

3.2 Persistent Object Protocol

User-defined classes inherit from <persistent-object>:

```
(defclass <person> [<persistent-object>]
  [[name :type <string>]
   [age :type <number>]])

(def alice (make <person> :name "Alice" :age 30))
(def older-alice (assoc alice :age 31)) ; Returns new instance
(:age alice) ; => 30 (unchanged)
(:age older-alice) ; => 31
```

defclass supports standard CLOS options (:type, :initarg, :initform, :reader, :writer). Slot values are stored in a persistent hash map, enabling $O(\log n)$ updates with structural sharing.

3.3 Collection Implementation

FOL wraps FSet [3] and Sycamore data structures for its collections:

- **Vectors** ([1 2 3]): 32-way branching trees supporting $O(\log_{32} n)$ random access
- **Maps** ({:a 1 :b 2}): Hash array mapped tries (HAMTs) with efficient key-value operations
- **Sets** ({1 2 3}): Weight-balanced binary trees for ordered iteration
- **Lists**: Persistent cons cells with standard $O(1)$ prepend and $O(n)$ access

A wrapper layer transparently converts between CL and FOL persistent objects.

3.4 Runtime Representation

Closures are persistent objects containing the function body, captured environment (an FSet map), and optional metadata. Immutable closed-over values enable safe sharing across threads.

Atoms are mutable references to persistent values (along with streams and regular-expression scanners, exceptions to immutability). swap! acquires a per-atom lock, applies the update function, and stores the result.

Garbage collection relies on Common Lisp's GC. Old versions are collected when unreferenced; structural sharing ensures only unshared portions are reclaimed.

3.5 Versioned Objects via Persistence

The persistent metaclass (Section 3.6) provides the foundation for a versioned object system. Because assoc returns a new object sharing structure with the old, version chains are cheap to construct:

```
(defclass <versioned> [<persistent-object>]
  [[version :initform 0]
   [previous :initform nil]])

(defn with-version [obj slot-name new-value]
  (-> obj
    (assoc slot-name new-value)
    (assoc :version (inc (:version obj)))
    (assoc :previous obj)))

(defn object-history [obj]
  (take-while some? (iterate :previous obj)))
```

Persistence makes the Memento pattern implicit: each assoc produces a snapshot at $O(\log n)$ cost with structural sharing, versus $O(n)$ deep copies traditionally. The MOP's slot-value-using-class redirection (Section 3.6) makes assoc on objects possible; Clojure maps support cheap assoc but lack metaclasses to encapsulate versioning as a reusable protocol.

3.6 MOP Protocol Adaptation

FOL's persistent metaclass adapts rather than replaces the MOP. Table 2 summarizes the protocol disposition.

Introspection protocols (class-slots, class-precedence-list, slot-definition-*) work unchanged because FOL's class *structure* is standard CLOS—only slot *storage* is redirected. The validate-superclass extension permits cross-metaclass inheritance (persistent classes may inherit from standard classes), but mixed hierarchies must

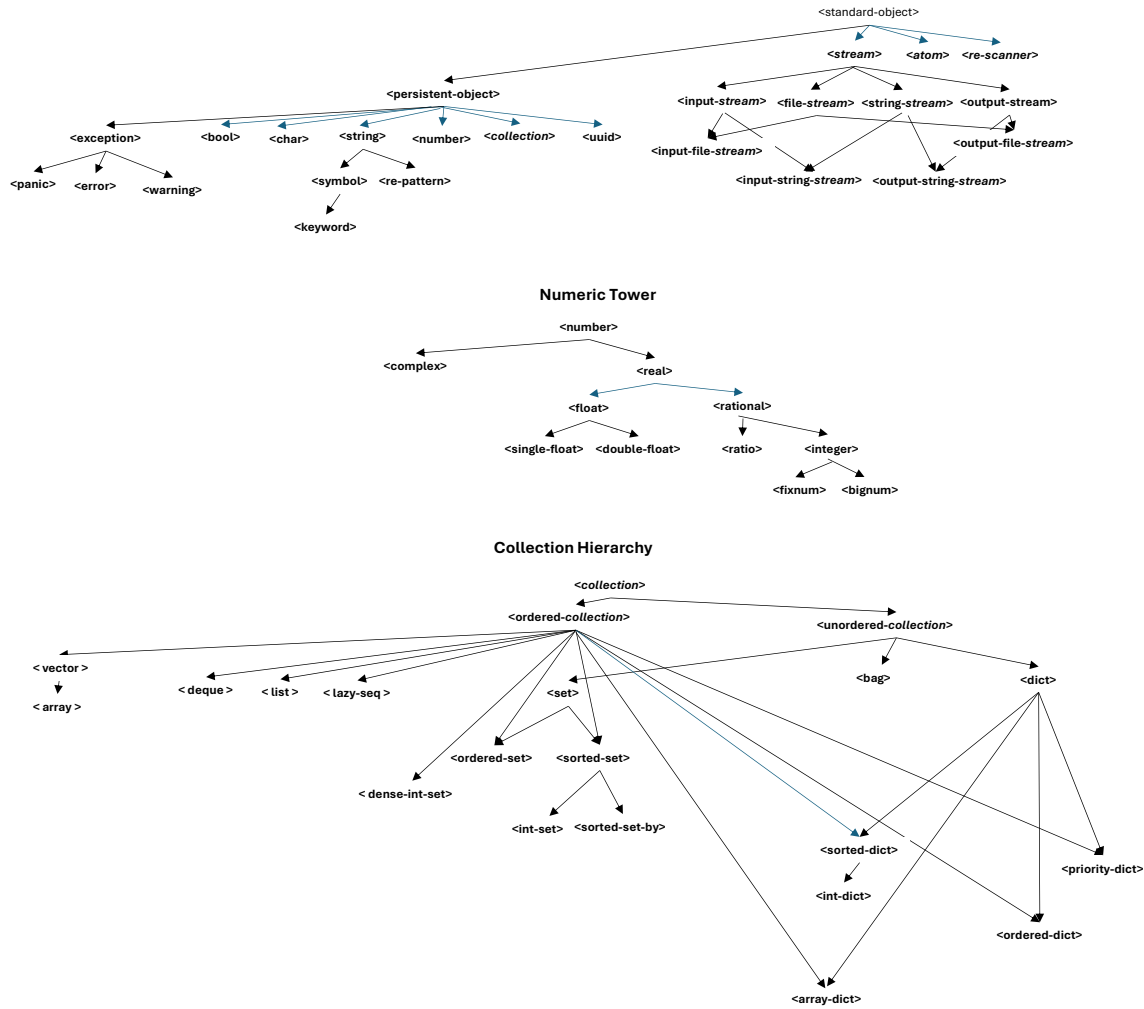


Figure 1: FOL Type Hierarchy

respect the invariant that persistent slots use FSet storage while inherited standard slots remain mutable—finalization signals an error if this invariant cannot be maintained. This adaptation requires the MOP: persistence cannot be implemented as a library because transparent slot access demands slot-value-using-class redirection at the metaclass level.

4 Features

4.1 Clojure Features

FOL implements key Clojure abstractions:

- **Transducers** [12]: Composable transformation pipelines separating the “what” (map, filter, take) from the “how” (eager, lazy, parallel).

- **Lazy sequences**: On-demand computation enabling infinite sequences and efficient streaming.
- **Thread-safe atoms**: Mutable references with per-atom locking for concurrent updates.
- **Sequence functions**: Over 100 functions including map, filter, reduce, take, drop, partition, group-by, and frequencies.
- **Macros**: Clojure-style defmacro with syntax-quote (```), unquote (`~`), and splicing unquote (`~@`); macros receive unevaluated forms. Example: `(defmacro unless [test & body] `(if (not ~test) (do ~@body)))`.
- **Error handling**: try/catch/throw forms providing basic exception handling via Common Lisp’s condition system.

Metadata, refs/STM, and agents are planned.

Table 2: MOP Protocol Adaptation

Protocol	Status	Rationale
slot-value-using-class	Redirected	Reads from FSet map
(setf slot-value-using-class)	Guarded	Blocks post-init writes
slot-boundp-using-class	Redirected	Checks FSet map membership
slot-makunbound-using-class	Blocked	Immutability invariant
validate-superclass	Extended	Cross-metaclass mixing
initialize-instance	Extended	Populates FSet map from initargs
update-instance-for-redefined-class	Omitted	Instances are immutable; lazy migration instead (Sec. 5.4)
change-class	Omitted	Violates immutability
<i>Class computation, method combination, allocation: standard CLOS behavior inherited unchanged.</i>		

4.2 Generic Function Integration

FOL’s generic functions dispatch by arity first, then by type predicates (`<vector>?`, `<dict>?`) and specificity within an arity group. Methods at different arities may coexist on the same generic:

```
(defgeneric process [x])
(defmethod process [(x <vector>)] (vec (map process x)))
(defmethod process [(x <dict>)] (update-vals x process))
(defmethod process [(x <number>)] (* x 2))

(process [1 { :a 2 :b 3 } [[4]]]) ; => [2 { :a 4 :b 6 } [[8]]]

;; Multi-clause methods with predicate dispatch
(defmethod process
  ([a (b (< 10))] (* 2 b))
  ([a (b <number>) (c <string>)] c)
  ([a b] (str a b)))

(defn is-buy-side-trade?
  ([([trade (-> (trade-side) (= :buy)))] t)
  ([trade] nil))
```

Figure 2 gives the formal grammar for FOL’s multi-pattern dispatch syntax.

Predicate semantics: The form `(symbol (fn arg0 arg1 ...))` evaluates `(fn symbol arg0 arg1 ...)`—the bound value is inserted as the first argument. This works with functions, macros, or special forms.

defn vs defgeneric/defmethod: `defn` defines a standalone multi-clause function with predicate dispatch and specificity ordering; it does not create a generic function. `defgeneric` declares a generic function; `defmethod` adds methods to it. Only `defmethod` accepts `:before/`, `:after/`, `:around` qualifiers. Both `defn` and `defmethod` support multi-clause predicate dispatch.

4.2.1 Method Combinations. Multi-clause `defmethod` forms expand to N separate method definitions. The `:before`, `:after`, and `:around` qualifiers apply per-clause; `call-next-method` follows CLOS semantics.

4.3 Predicate Specializers

FOL extends pattern matching with predicate specializers. The syntax `(var (fn arg0 arg1 ...))` applies the predicate to the bound value at runtime.

4.3.1 Predicate Examples. Predicates work with any function—equality, comparison, type tests, or user-defined. In the first example

below, all clauses are predicates (same category), so definition order determines dispatch; the second example mixes type predicates and a catch-all, where cross-category specificity applies:

```
;; Range checking with comparison predicates
(defn age-group
  ([ (age (< 13))] :child)
  ([ (age (< 20))] :teen)
  ([ (age (< 65))] :adult)
  ([age] :senior))

;; Type predicates and user-defined predicates
(defn process-value
  ([ (x (<number>?))] (* x 2))
  ([ (x (<string>?))] (str x x))
  ([ (x (<vector>?))] (vec (map process-value x)))
  ([x] x))
```

4.3.2 Predicate Specificity. FOL orders patterns by *constraint strength*—how narrowly a pattern constrains its accepted values. More specific patterns are tested first, ensuring predictable dispatch regardless of definition order:

Table 3: Predicate Specificity Levels

Level	Pattern Type	Constraint	Example
3	Predicate	Single value/narrow range	<code>(n (= 5))</code>
2	Type	All instances of a type	<code>(x <number>)</code>
1	Destructuring	Structural shape only	<code>[a b c]</code>
0	Any	Universal match	<code>x, _</code>

This ordering is a deliberate heuristic chosen for predictability rather than full logical subsumption, which is undecidable in general [7]. The intuition is that a predicate like `(= 5)` accepts strictly fewer values than `<number>`, which accepts fewer than an untyped destructuring pattern. The boundary between type and predicate categories is conventional: type predicates like `<number>?` are recognized syntactically and assigned level 2, while other predicates (including user-defined type tests) receive level 3. This means `(x (<number>?))` and `(x <number>)` accept the same values but occupy different specificity levels if mixed: a clause using `<number>?` (level 3) will shadow one using `<number>` (level 2) for numeric arguments, even though both match the same set. In practice this is benign—programmers would use one form

<code>defn</code>	<code>::= (defn name (single-clause multi-clause))</code>	
<code>defgeneric</code>	<code>::= (defgeneric name (single-pattern multi-pattern))</code>	
<code>defmethod</code>	<code>::= (defmethod name qualifier[?] (single-clause multi-clause))</code>	
<code>fn</code>	<code>::= (fn (single-clause multi-clause))</code>	
<code>single-clause</code>	<code>::= pattern body⁺</code>	
<code>multi-clause</code>	<code>::= (pattern body⁺)⁺</code>	
<code>single-pattern</code>	<code>::= pattern</code>	
<code>multi-pattern</code>	<code>::= pattern⁺</code>	
<code>pattern</code>	<code>::= [param*] [param* & rest-param]</code>	
<code>param</code>	<code>::= symbol</code>	(simple binding)
	<code> (symbol type)</code>	(type specifier)
	<code> (symbol (fn-name arg*))</code>	(predicate specifier)
	<code> destructure</code>	(nested destructuring)
<code>destructure</code>	<code>::= [param*]</code>	(sequential)
	<code> { :keys [key-spec*] }</code>	(map)
<code>key-spec</code>	<code>::= symbol (symbol type) (symbol pred)</code>	
<code>type</code>	<code>::= <type-name></code>	
<code>qualifier</code>	<code>::= :before :after :around</code>	

Figure 2: FOL Multi-Pattern Dispatch Grammar (simplified)

or the other, not both—but it is a pragmatic tradeoff we accept for uniform, predictable dispatch. Within a category, definition order is preserved, giving programmers explicit control over tie-breaking.

For sequence patterns, specificity extends to element-level comparisons: $[(x (< 0)) y]$ is more specific than $[(x < \text{number}>) y]$ because its first element has higher specificity.

4.3.3 Formal Specificity Ordering. We formalize the specificity relation $\prec$ using inference rules. Let $\mathcal{L}(p) \in \{0, 1, 2, 3\}$ be the specificity level of pattern p .

SPEC-LEVEL $\mathcal{L}(p_1) > \mathcal{L}(p_2)$	SPEC-SEQ-HEAD $p_1 \prec q_1 \quad n \geq 1$
$p_1 \prec p_2$	$[p_1 \dots p_n] \prec [q_1 \dots q_n]$
SPEC-SEQ-TAIL $\mathcal{L}(p_1) = \mathcal{L}(q_1)$	$[p_2 \dots] \prec [q_2 \dots]$
	$[p_1 \dots p_n] \prec [q_1 \dots q_n]$

The operational dispatch order $\prec_{\text{disp}}$ is a total order that resolves ties (where neither $p_1 \prec p_2$ nor $p_2 \prec p_1$) using definition order. Let $\text{idx}(p)$ denote the definition index of the method clause containing p .

$$p_1 \prec_{\text{disp}} p_2 \iff p_1 \prec p_2 \vee (\neg(p_2 \prec p_1) \wedge \text{idx}(p_1) < \text{idx}(p_2))$$

Patterns of different arity are incomparable under $\prec$; dispatch selects the arity group first, then applies $\prec_{\text{disp}}$ within it. This lexicographic combination ensures deterministic dispatch: arity-first, then specificity, then definition-order.

4.3.4 Dispatch Semantics and Type Checking. Predicate specializers also work in anonymous `fn` forms but not in macro parameter lists. Dispatch is $O(N)$ in the number of patterns at a given arity, with patterns pre-sorted by specificity. FOL’s type system is entirely dynamic—type specializers are checked at runtime. The interpreter

does not warn about unreachable clauses; a future compiler could detect shadowing statically.

4.3.5 Soundness Properties. FOL’s dispatch maintains three invariants: (1) **exhaustive matching**—unmatched calls signal an error no-matching-clause rather than silently failing; (2) **deterministic dispatch**—the total order $\prec_{\text{disp}}$ ensures exactly one clause matches; (3) **type predicate consistency**—type specializers use the same predicates as explicit checks. FOL does *not* provide static type soundness; errors are detected at runtime only.

5 Synergy in Practice

This section demonstrates patterns that are *more natural or more concise* in FOL than in either Clojure or standard CLOS alone, because they exploit the combination of persistent data, CLOS method combinations, and predicate dispatch. We focus on patterns that *require* multiple features working together—not merely features used in proximity but features whose interactions produce capabilities unavailable in either parent language. Full source for all examples is available in the repository: FOL code in `fol-code/`, Common Lisp equivalents in `lisp-code/`, and Clojure equivalents in `clojure-code/`.

5.1 Declarative Dispatch: Trade Compliance

We implement a trade compliance validator in FOL and plain Common Lisp (repository files `compliance.fol`, `compliance.lisp`). The FOL version uses predicate dispatch with automatic specificity ordering; the CL version uses a manual `cond` cascade with hand-ordered predicates. Table 4 compares the implementations (non-blank, non-comment lines).

The savings are modest (17%) because this example exercises predicate dispatch and module syntax but not persistence or method combinations—it demonstrates a single-feature advantage (predicate dispatch) rather than multi-feature synergy. The predicate

Table 4: Code Comparison—Trade Compliance

Category	FOL	CL	Savings
Class definition	5	8	38%
Predicates / logic	16	13	−23%
Dispatch / validation	15	15	0%
Module boilerplate	2	10	80%
Total	38	46	17%

category is *larger* in FOL because predicate-dispatch patterns require explicit true/nil branches. The advantage here is *correctness*, not brevity: FOL’s specificity ordering guarantees that more-constrained rules fire first regardless of definition order. Reordering or adding rules cannot silently change which clause matches—a class of maintenance bug that plagues cond cascades in production.

5.2 Event Sourcing with Method Combinations

FOL enables event sourcing by combining `:around` methods with persistent event logs. The key mechanism is a single `:around` method that intercepts all command dispatches:

```
(defmethod apply-command :around [agg cmd]
  (bind [result (call-next-method)
        event {:command cmd :timestamp (now)}]
    (assoc result :events
      (conj (:events result) event))))
```

This `:around` method applies automatically to every `apply-command` invocation, including future methods added after deployment. The Clojure equivalent uses `defmulti/defmethod` but lacks `:around` methods, so event logging requires a manual wrapper that callers must remember to invoke. Table 5 compares the two (see repository for full source).

Table 5: Code Comparison—Event Sourcing

Category	FOL	Clj	Notes
Class / factory	2	2	Map literal
Event logging	4	4	<code>:around</code> vs wrapper
Command methods	7	5	Pred. vs key dispatch
Replay	2	2	Identical
Total	15	14	

Line counts are nearly identical, but the two implementations differ qualitatively. FOL’s `:around` method is *declarative*—it applies automatically to all invocations, including future methods; Clojure’s wrapper requires callers to remember to use `apply-command-logged`, and calling the multimethod directly silently skips logging. Persistence contributes a second guarantee: because `assoc` and `conj` produce new immutable values, the event log is append-only by construction—no code path can retroactively alter a recorded event, which is the core invariant event sourcing requires. In mutable CLOS, both the `:around` interception and defensive copying of the event list would be needed to achieve the same guarantee.

5.3 Observable Slots via MOP and Predicate Dispatch

This example combines persistent functional updates with predicate dispatch to build a change-notification system (repository file `observable.fol`). When a slot is updated, the protocol produces both a new object (via persistent `assoc`) and a change event; predicate dispatch routes notifications by content—e.g., detecting reading spikes (>50 unit jump) or critical status transitions. The CL equivalent (`observable.lisp`) requires a manual copy function, a separate `defstruct` for change events, and a hand-coded cond cascade. Table 6 compares the implementations.

Table 6: Code Comparison—Observable Slots

Category	FOL	CL	Savings
Class / boilerplate	6	23	74%
Update protocol	7	9	22%
Change dispatch	18	20	10%
Total	31	52	40%

The 40% reduction comes primarily from eliminating boilerplate: FOL needs no manual copy function (persistent `assoc` shares structure), no separate `defstruct` for change events (persistent maps suffice), and no package ceremony. This example best illustrates three-way synergy: persistence provides safe functional updates that produce change events without corrupting the original; predicate dispatch routes those events by content (e.g., reading spikes, critical transitions); and generic functions with `defmethod` enable new notification rules to be added without modifying the existing dispatch. None of these features alone suffices—Clojure has persistence but no method dispatch for extensible routing; CLOS has dispatch but requires manual deep copies to produce safe change events.

5.4 Lazy Schema Evolution

Standard CLOS handles class redefinition by lazily updating existing instances via `update-instance-for-redefined-class`: each instance is mutated in place the next time it is accessed. This protocol assumes mutable instances—an assumption FOL deliberately violates.

FOL’s persistent metaclass resolves this tension through what amounts to *lazy schema evolution*. Class definitions remain mutable (redefining a class updates the class object normally), but existing instances are frozen snapshots whose `%persistent-storage` FSet maps are never modified. The MOP’s `slot-value-using-class` method provides graceful degradation: when accessing a slot that was added by a redefinition but is absent from an old instance’s storage map, the method falls back to evaluating the slot’s `:initform` if one exists, or signals `slot-unbound` otherwise. Slots removed by redefinition remain in the storage map but become inaccessible (no slot definition routes to them).

Crucially, functional updates perform implicit migration of already existing objects. `set-pslot-value` calls `(class-of object)` to obtain the *current* (redefined) class, allocates a fresh instance of that class, and copies the old storage map forward. The resulting

object is an instance of the new schema carrying the old data—new slots resolve via `initforms`, removed slots persist harmlessly in the map.

Persistence makes this *safe* in a specific sense: old instances are never corrupted and remain valid snapshots of the prior schema; in mutable CLOS, redefinition silently mutates every reachable instance. The tradeoff is that CLOS’s `update-instance-for-redefined-class` provides an explicit migration hook where the programmer can transform slot values, while FOL’s lazy approach silently returns `initform` defaults for new slots on old instances—a form of graceful degradation that could mask schema mismatches if the programmer is not aware. This lazy migration strategy is, to our knowledge, a novel adaptation of the MOP for immutable object systems—it trades explicit migration control for corruption safety and zero-downtime operation.

6 Evaluation

6.1 Comparison with Related Languages

Table 7: Built-in Feature Comparison

Feature	FOL	Clojure	CL
Persistent objects	✓	✓	×
CLOS/MOP	✓	×	✓
Method combinations	✓	×	✓
Transducers	✓	✓	×
Lazy seqs	✓	✓	×
Predicate dispatch	✓	~ ^a	× ^b
Macros	✓	✓	✓

^adefault dispatches on an arbitrary fn but lacks per-parameter specializers and specificity ordering.

^bEQL specializers only; `optima/trivia` provide pattern matching as libraries.

FOL uniquely combines Clojure’s functional features with Common Lisp’s object system. Persistent structures incur $O(\log n)$ vs $O(1)$ for mutable operations, but high branching factors (32–64) keep constant factors small.

6.2 Benchmark Methodology

All measurements were taken on SBCL 2.6.0 (Windows 11, AMD Ryzen 9 5900X, 64 GB RAM). Performance benchmarks report the mean of 100 runs (coefficient of variation <2%); memory benchmarks report single allocations via `sb-ext:dynamic-space-usage`. We compare against native Common Lisp rather than Clojure: SBCL compiles to native code, providing a strict upper bound on overhead, and since FOL’s implementation is Common Lisp, this isolates abstraction cost. A Clojure comparison would conflate language design with JVM runtime differences (startup, JIT warmup, GC pauses).

6.3 Memory Benchmarks

FSet sequences incur ~1.42x memory overhead at scale, purchasing lock-free access and cheap snapshots.

Small objects (2–5 slots) have modest overhead (1.1–1.4x); larger objects (50–100 slots) grow to 3–4x.

Table 8: Vector Memory Comparison

N	CL (bytes)	FSet (bytes)	Ratio
10,000	131,008	327,472	2.50x
100,000	2,372,112	3,372,960	1.42x
1,000,000	23,982,992	33,959,040	1.42x

Table 9: Per-Slot Memory Overhead (10,000 instances)

Slots	CL bytes/inst	FSet bytes/inst	Ratio
2	216.2	236.0	1.09x
5	491.0	701.0	1.43x
10	923.0	1,707.0	1.85x
20	1,808.0	3,934.0	2.18x
50	4,455.0	13,565.0	3.04x
100	8,870.0	34,970.0	3.94x

Structural Sharing: Retaining 1000 versions of a 1000-element vector:

Table 10: Structural Sharing Memory (1000 versions, 1000-element vector)

Approach	Memory (bytes)	Per-version
CL (full copy each mutation)	24,000,000	24,000
FSet (structural sharing)	1,200,000	1,200
Sharing ratio	20x savings	

6.4 Performance Benchmarks

Table 11: Map Performance Comparison

N	CL (sec)	FOL (sec)	Ratio
9	0.000001	0.000263	263x
99	0.000004	0.003023	756x
999	0.000075	0.025619	342x
9,999	0.000669	0.341401	510x

The 300–500x gap reflects interpretation overhead; FSet compiled to native code achieves 2–3x of mutable. Predicate and conditional dispatch show identical performance—interpreter overhead dominates both.

6.5 Isolating Persistence from Interpretation

To separate the cost of persistence from the cost of interpretation, we benchmarked FSet operations compiled directly to native code (no FOL interpreter), averaged over 100 runs. Sequential operations (map, reduce) incur 1.1–2.7x overhead; random updates incur 21–22x ($O(\log n)$ vs $O(1)$); dictionary lookups and inserts incur 6–39x. Comparing these figures to the 300–500x overhead in the Map Performance table, we conclude that **interpretation accounts for**

the vast majority of overhead, while persistence adds only 1–3x for sequential operations. Compilation is therefore the primary optimization target.

6.6 Preliminary Transpiler Results

To validate the claim that compilation is the primary optimization target, we transpiled the compliance example (Section 5.1) from FOL source to Common Lisp using a preliminary, partial transpiler. The transpiler handles `defclass`, `defgeneric/defmethod` with multi-clause predicate dispatch, `defn` with predicate specializers, and threading macros. The core algorithm (class definitions, predicate functions, and dispatch logic—approximately 85% of the benchmarked code) was produced by the transpiler; only the surrounding package boilerplate was written by hand.

Naïve emission: Multi-clause functions initially transpile to `(&rest args)` lambdas with a `cond` cascade that checks arity via `(length args)` and extracts arguments via `(nth 0 args)` at each clause. This incurs per-call overhead from rest-argument consing, list-length computation, and list indexing.

Fixed-arity optimization: When all clauses of a multi-clause function share the same arity and none use rest parameters, the transpiler detects this uniform-arity case and emits a fixed-parameter lambda `(lambda (a0 a1 ...) (cond ...))` with direct variable references, eliminating rest-argument consing, length computation, and list indexing entirely. In the compliance module, all six multi-clause functions have uniform arity, so the optimization applies throughout.

We benchmarked three versions of the compliance module—hand-written CL, unoptimized transpiled, and optimized transpiled—all compiled by SBCL with `(speed 3)`. Each version validates four test trades covering all dispatch paths (approved, rejected, manual-review, warning); we report the best of 5 runs of 10,000 iterations each to minimize timing variance.

Table 12: Transpiled FOL vs. Hand-Written CL (Compliance Module)

Version	Per-call	Ratio
Hand-written CL	0.13 μ s	1.0x
Transpiled FOL (unoptimized)	0.18 μ s	1.4x
Transpiled FOL (fixed-arity)	0.12 μ s	0.93x

The unoptimized transpiler already produces code within 1.4x of hand-written CL, reflecting the modest cost of `&rest` dispatch under SBCL’s `(speed 3)` optimization. The fixed-arity optimization eliminates this remaining overhead, reaching parity with hand-written code (0.93x—within measurement noise). The transpiled code performs the same logical operations as the hand-written version; the only difference was the dispatch mechanism, which the optimization removes. Comparing against the 300–500x interpreted overhead from Section 6.4 (a different workload), these results confirm that **compilation eliminates dispatch overhead entirely**, and that interpretation rather than language design accounts for FOL’s current performance gap.

6.7 Threats to Validity

All measurements except the transpiler benchmark (Section 6.6) compare FOL’s interpreter against SBCL’s native compiler, so reported ratios reflect interpretation overhead, not inherent design costs; the transpiler benchmark compares three SBCL-compiled programs (hand-written, unoptimized transpiled, and optimized transpiled). We do not benchmark against Clojure (see Section 6.2). Memory measurements via `sb-ext:dynamic-space-usage` may include GC artifacts. Code density comparisons use three small, purpose-built examples (38–52 lines); results may differ for larger programs where cross-cutting concerns interact. As a single-author project, FOL has not yet undergone independent code review; the full source is publicly available to facilitate external evaluation.

7 Limitations and Future Work

FOL is primarily interpreter-based; a preliminary transpiler exists (Section 6.6) but covers only a subset of the language. No static analysis exists—type errors and unreachable clauses are detected at runtime. Memory overhead grows with object size (1.1x for 2 slots, 4x for 100 slots). Random updates incur 21x overhead versus $O(1)$ mutation. Class redefinition does not migrate existing instances (Section 5.4); old objects retain their original slot data, with new slots resolved lazily via `initforms`. Metadata, `refs/STM`, and agents are not yet implemented.

Planned extensions include **completing the native compiler** (the primary optimization target—the preliminary transpiler already reaches parity with hand-written CL on dispatch-intensive code), **abstract and sealed classes** for exhaustive pattern matching, **parallel collections** following Scala’s model, full **condition and restart support** (exposing `handler-bind`, `restart-case`, and interactive restarts beyond the current `try/catch`), **STM refs** following Clojure’s `ref/dosync` model, and **channel-based concurrency** via CSP-style go blocks.

8 Related Work

Clojure [10] pioneered persistent data structures in Lisp. Clojure’s *protocols* provide type-based polymorphism (single dispatch on the first argument’s type) as a lightweight alternative to CLOS generic functions. FOL adopts Clojure’s sequence abstraction and transducers but uses CLOS generic functions with multiple dispatch and method combinations—gaining `:before/after/around` qualifiers and MOP introspection at the cost of Clojure’s protocol performance optimizations.

Common Lisp’s CLOS [2] and MOP [13] provide FOL’s object system foundation, extended with persistent slots.

Dylan [16] influenced naming conventions (`<type>` notation) and module semantics.

Persistent data structures build on Okasaki [14]; `FSet` [3] provides production-quality collections for Common Lisp. FOL’s functional update pattern is related to **lenses** in the Haskell tradition; FOL does not yet provide a lens abstraction. **Clojure’s** `defrecord`, Scala case classes, and Kotlin data classes all provide value-oriented objects, but none integrate with a MOP—FOL is unique in combining persistent storage with metaclass-level extensibility.

8.1 Predicate Dispatch and Pattern Matching

Ernst et al. [7] introduced predicate dispatch with logical implication for specificity. Chambers and Chen [4] demonstrated efficient implementation strategies using dispatch DAGs, achieving performance competitive with hand-written type-case dispatch. **Filtered Dispatch** [15] extended CLOS with predicate-based method selection using explicit priorities. FOL’s category-based specificity with first-match tie-breaking provides a middle ground between logical implication (expressive but undecidable) and explicit priorities (simple but error-prone). **Scala’s extractors** [6] and **Racket’s match expanders** provide extensible pattern matching; **OCaml’s polymorphic variants** [9] offer extensible sum types. FOL differs: predicates are arbitrary functions rather than structural extractors, and specificity is determined by category rather than type inclusion.

8.2 Multiple Dispatch and Typed Functional Lisps

Julia [1] uses multiple dispatch but is purely type-based, lacking predicate specializers, method combinations, and MOP introspection. **Coaltion** [17] adds Hindley-Milner types to CL—complementary to FOL (static safety vs. persistence). **Typed Racket** [21] adds gradual typing; FOL’s type specializers serve as dispatch mechanisms rather than correctness guarantees. **Shen** [20] and **Carp** [19] offer pattern matching and static typing respectively, but lack multiple dispatch with method combinations. **Racket’s class system** [8] provides method combinations but separates classes from match; FOL unifies pattern matching and method dispatch. **Elixir** combines pattern matching with protocols on the BEAM VM but lacks multiple dispatch and method combinations.

9 Conclusion

FOL demonstrates that persistent objects and CLOS are synergistic rather than competing paradigms. Their combination yields versioned objects via persistent snapshots, declarative event sourcing via method combinations over persistent logs, and observable slots via predicate dispatch over change events—patterns that require multiple features working together and are more natural than in either parent language alone. Lazy schema evolution (Section 5.4) further illustrates how immutability transforms a traditionally problematic MOP protocol into a corruption-safe migration strategy, albeit at the cost of CLOS’s explicit migration hooks.

Benchmarks confirm that interpretation, not persistence, dominates current overhead (300–500x vs 1–3x). A preliminary transpiler to Common Lisp with a single optimization—fixed-arity dispatch—reaches parity with hand-written code (0.93x) on the compliance

benchmark, eliminating the dispatch overhead entirely. This demonstrates that compilation, not language design, is the sole performance bottleneck, and that a complete compiler can retain immutability’s correctness and concurrency benefits at native speed.

The complete implementation—385 functions, 6,333 tests, and ~18,000 lines of bootstrap code—is available at: <https://github.com/frankadrian314159/fol>

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